

MATH 301: Homework 5

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1 Section 5.3 - 2

Let $I := [a, b]$ and let $f : I \rightarrow \mathbb{R}$ and $g : I \rightarrow \mathbb{R}$ be continuous functions on I . Show that the set $E := \{x \in I : f(x) = g(x)\}$ has the property that if $(x_n) \subseteq E$ and $x_n \rightarrow x_0$, then $x_0 \in E$.

Statement (1): Observe $\forall n \in \mathbb{N}, x_n \in E \implies f(x_n) = g(x_n)$.

Statement (2): Observe $\forall x \in E, x \in I$. Naturally then as I is a closed bounded interval and $E \subseteq I$. E must also be closed and bounded; else there must exist some element of E that escapes the bounds which would invalidate the fact that all elements of E come from I .

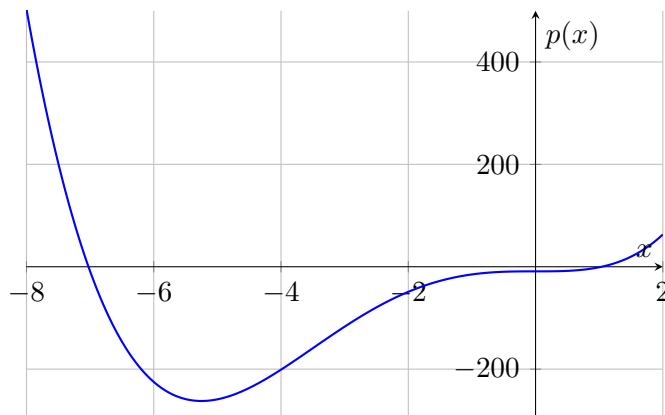
As f, g are continuous in the bounded interval $[a, b]$ the *sequential criterion of continuity* must hold for values of E . Thus, $\lim_{n \rightarrow \infty} f(x_n) = f(x_0)$ and $\lim_{n \rightarrow \infty} g(x_n) = g(x_0)$

As this is true we are left to conclude $f(x_0) = g(x_0)$. Therefore $x_0 \in E$.

2 Section 5.3 - 5

Show that the polynomial $p(x) := x^4 + 7x^3 - 9$ has at least two real roots. Use a calculator to locate these roots to within two decimal places.

Visualization $p(x)$



Invoke **IVT**.

$$p(1) = 1 + 7 - 9 = -1$$

$$p(2) = 16 + 7(16) - 9 > 0$$

$$p(-8) = -8^4 + 7(-8)^3 - 9 > 0$$

As the function is a continuous and there exists $p(-8) < p(1) < p(2)$ where $0 < p(-8), p(2)$ and $p(1) < 0$. There must exist some $c_1, c_2 \in \mathbb{R}$ st $p(c_1) = 0 = p(c_2)$.

Approximation:

3 Section 5.3 - 11

Let $I := [a, b]$, let $f : I \rightarrow \mathbb{R}$ be continuous on I , and assume that $f(a) < 0, f(b) > 0$. Let $W := \{x \in I : f(x) < 0\}$, and let $w := \sup W$. Prove that $f(w) = 0$. (This provides an alternative proof of Theorem 5.3.5.)

Note $f(a) < 0 < f(b)$ and $f(x)$ is continuous on $[a, b]$. By IVT $\exists c \in \mathbb{R}$ st $f(c) = 0$.

By W 's construction, $\forall w \in W, f(w) < 0$. As $a < b$ and $0 < f(b)$ then it means that b is an upper bound of W . This means $\exists \sup W := z$ where $z - \epsilon$ is not an upper bound.

By the definition of the supremum, for any $n \in \mathbb{N}$, $w - \frac{1}{n}$ is not an upper bound of W . Thus, there exists some $x_n \in W$ such that $w - \frac{1}{n} < x_n \leq w$. This creates a sequence $(x_n) \subseteq W$ where $\lim_{n \rightarrow \infty} x_n = w$. Since $x_n \in W, f(x_n) < 0$. By the continuity of $f, f(w) = \lim_{n \rightarrow \infty} f(x_n) \leq 0$. Next, construct a sequence (y_n) defined by $y_n = w + \frac{1}{n}$. For sufficiently large $n, y_n \in I$. Since $y_n > w = \sup W$, it follows that $y_n \notin W$, meaning $f(y_n) \geq 0$. By continuity, $f(w) = \lim_{n \rightarrow \infty} f(y_n) \geq 0$. Since $f(w) \leq 0$ and $f(w) \geq 0$, we conclude $f(w) = 0$.

Root 1: $x \approx 1.04$ (since $p(1.04) \approx -0.07$ and $p(1.05) \approx 0.92$)

Root 2: $x \approx -7.02$ (since $p(-7) = -9$ and $p(-8) = 503$)

4 Section 5.4 - 2

Show that the function $f(x) := 1/x^2$ is uniformly continuous on $A := [1, \infty)$, but that it is not uniformly continuous on $B := (0, \infty)$.

Consider $x, y \in A \implies 1 \leq x, y$

Observe:

$$\begin{aligned} |1/x^2 - 1/y^2| &= \left| \frac{(x+y)(x-y)}{x^2y^2} \right| \\ &= \left| \frac{x+y}{x^2y^2} \right| |x-y| \\ &= \left[\frac{1}{xy^2} + \frac{1}{x^2y} \right] |x-y| \end{aligned}$$

Consider ordering:

$$\begin{aligned} x \geq 1 &\implies 1/x \leq 1 \implies 1/x^2 \leq 1^2 \\ 1/x^2 \leq 1^2 \wedge 1/y \leq 1 &\implies 1/(x^2y) \leq 1 \end{aligned}$$

Note this is the same for $1/(xy^2) \leq 1$. Therefore, $1/(xy^2) + 1/(x^2y) \leq 2$.

We can use this to bound the following:

$$|1/x^2 - 1/y^2| = \left[\frac{1}{xy^2} + \frac{1}{x^2y} \right] |x-y| \leq 2|x-y|$$

Then we can conclude that $f(x)$ is continuous on A .

If $x_n := \frac{1}{n}$ and $y_n := \frac{1}{(n+1)}$. Then $|x_n - y_n| \rightarrow 0$ However $\forall n \in \mathbb{N}$, $|f(x_n) - f(y_n)| = 2n + 1 \geq 1$ so $f(x)$ is not uniformly continuous on $(0, \infty)$

5 Section 5.4 - 4

Show that the function $f(x) := 1/(1 + x^2)$ for $x \in \mathbb{R}$ is uniformly continuous on $\mathbb{R}$.

6 Section 5.4 - 5

Show that if f and g are uniformly continuous on a subset A of $\mathbb{R}$, then $f + g$ is uniformly continuous on A .

Consider:

$$\begin{aligned} & |(f(x) + g(x)) - (f(y) + g(y))| \\ &= |(f(x) + f(y)) - (g(x) + g(y))| \\ &\leq |(f(x) - f(y))| + |(g(x) - g(y))| \end{aligned}$$

Which holds as we are given $f(x), g(x)$ is uniformly continuous on $\mathbb{R}$.

7 Section 5.4 - 6

Show that if f and g are uniformly continuous on $A \subseteq \mathbb{R}$ and both bounded on A , then their product fg is uniformly continuous on A .

Since f and g are bounded on A , there exist constants $M_f, M_g \geq 0$ such that

$$|f(x)| \leq M_f \quad \text{and} \quad |g(x)| \leq M_g \quad \text{for all } x \in A.$$

To prove that fg is uniformly continuous on A , let $\varepsilon > 0$ be given. Since f is uniformly continuous on A , there exists $\delta_1 > 0$ such that

$$|x - y| < \delta_1 \implies |f(x) - f(y)| < \frac{\varepsilon}{2(M_g + 1)} \quad (x, y \in A).$$

Similarly, since g is uniformly continuous on A , there exists $\delta_2 > 0$ such that

$$|x - y| < \delta_2 \implies |g(x) - g(y)| < \frac{\varepsilon}{2(M_f + 1)} \quad (x, y \in A).$$

Let $\delta = \min\{\delta_1, \delta_2\}$. If $x, y \in A$ and $|x - y| < \delta$, then

$$\begin{aligned} |f(x)g(x) - f(y)g(y)| &= |f(x)g(x) - f(x)g(y) + f(x)g(y) - f(y)g(y)| \\ &\leq |f(x)| |g(x) - g(y)| + |g(y)| |f(x) - f(y)| \\ &\leq M_f |g(x) - g(y)| + M_g |f(x) - f(y)| \\ &< M_f \cdot \frac{\varepsilon}{2(M_f + 1)} + M_g \cdot \frac{\varepsilon}{2(M_g + 1)} \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

Therefore, fg is uniformly continuous on A .

8 Section 5.4 - 9

If f is uniformly continuous on $A \subseteq \mathbb{R}$, and $|f(x)| \geq k > 0$ for all $x \in A$, show that $1/f$ is uniformly continuous on A .

Let $x, y \in A$

$$|f(x)| \geq k > 0 \implies 0 < \frac{1}{|f(x)|} \leq \frac{1}{k}$$

$$|f(y)| \geq k > 0 \implies 0 < \frac{1}{|f(y)|} \leq \frac{1}{k}$$

We can construct the following:

$$\left| \frac{1}{f(x)} - \frac{1}{f(y)} \right| \leq \frac{1}{k^2} |f(x) - f(y)|$$

Let $\epsilon > 0$. Because f is uniformly continuous on A , there exists a $\delta > 0$ such that for all $x, y \in A$ with $|x - y| < \delta$, we have $|f(x) - f(y)| < k^2\epsilon$. Therefore, $|x - y| < \delta \implies \left| \frac{1}{f(x)} - \frac{1}{f(y)} \right| \leq \frac{1}{k^2}(k^2\epsilon) = \epsilon$. Thus, $1/f$ is uniformly continuous on A .